

Max Schulz

Email: max.schulz@uni-luebeck.de | Phone: 4945131013605 | Web: mxschlz.github.io | Address: Maria-Goeppert-Straße 9a, 23562 Lübeck

Education

- 2020 - 2024** **M. Sc. Biology**
University Leipzig, Germany
- Thesis in acoustic and computational neuroscience, "AI-Supported Evidence For Spectro-Temporal Primacy in Auditory Numerosity Judgment"
 - Tutoring how to prepare, conduct and analyze psychoacoustic electroencephalogram (EEG) experiments with Python
 - Implementation of custom-written platform which enables asynchronous and simultaneous operation of multiple devices (cameras and processors mainly) via a Python interface (AG Neurobiology)
 - Development of a custom analysis tool which enables EEG-analysis in MNE-Python (AG Neurobiology)
- 2017 - 2021** **B. Sc. Biology**
University Leipzig
- Bachelor Thesis in auditory neuroscience, "Measuring Auditory Tuning Curves With EEG"
- 2016 - 2017** **B. Sc. Biology**
Justus-Liebig University Gießen

Experience

- 2024 - now** **PhD Candidate**
University of Lübeck, Germany
- Researching subprocesses of auditory selective attention, specifically delineating target enhancement from distractor capture and suppression.
 - Utilizing EEG and webcam-based eye and head tracking to study neuro-cognitive mechanisms in young and older adults and infants.
 - Supervised by Dr. Malte Wöstmann and Prof. Dr. Jonas Obleser in the Auditory Cognition Lab.
- 2024** **Student Assistant**
Max-Planck Institute for Human Cognitive and Brain Sciences
- Assisting in the implementation of electrical cortical field simulations for optimization of transcranial magnetic stimulation approaches with PyNibs (Prof. Dr. Gesa Hartwigsen).
- 2021 - 2024** **Student Assistant**
Max-Planck Institute for Human Cognitive and Brain Sciences
- Assisting in study preparation, conduction, and analysis in the Psychology Department (Prof. Dr. Christian Doeller).
 - Conducting behavioral, eye-tracking, virtual-reality, and fMRI studies.

Open Source Projects

- 2020 - now** **mxschlz**
My GitHub repository, mostly containing Python containers for experiment and psychological analysis scripts.

Academic Interests

- Selective Attention
- Auditory Scene Analysis
- Electroencephalography (EEG)

Talks & Symposia

June 2026	Symposium Chair & Speaker <i>Psychologie und Gehirn (PuG) 2026, Heidelberg, Germany</i> • {'Organized and chaired the symposium "Dynamics of Enhancement and Suppression': 'Integrating Neural and Behavioral Perspectives on Selective Attention'"}
March 2026	Symposium Speaker <i>Tagung experimentell arbeitender Psychologen (TeaP) 2026, Tübingen, Germany</i> • Held a talk titled "Behavioral and neural signatures of extracting regularities from relevant and irrelevant sequences of sound"
Jan. 2026	Invited Talk <i>Marlene Meyer's Lab, Donders Institute, Nijmegen, Netherlands</i> • Presented research on the infant OCAPI (Objective Coding of Attention Pipeline for Infants) project

Teaching

June 2026	EEG Workshop Instructor <i>HörHanse Tage 2026, Lübeck, Germany</i> • Designed and conducted an interactive workshop on EEG application, setup, and analysis pipeline for ~50 participants
Occasional	Seminar Instructor (Statistics & EEG) <i>University of Lübeck & University of Leipzig, Germany</i> • Instructed student seminars on statistical methods and experimental EEG application (preprocessing, design, and data analysis)
2020 - 2024	Student Tutor (Psychoacoustic EEG) <i>University of Leipzig, Germany</i> • Tutored students in preparing, conducting, and analyzing psychoacoustic EEG experiments using Python and PsychoPy

Skills

Programming	• Python: PsychoPy, MNE-Python, PyNibs, numpy, scipy • Bash / Git • R
Research & Methods	• Electroencephalography (EEG) • Webcam-based Eye & Head Tracking • Psychoacoustics • Virtual Reality (VR)